LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Matter in Our Surroundings

Characteristics of particles of matter

- Particles of matter have spaces between them, are continuously moving, and attract each other.
- Diffusion (e.g. the smell of hot food spreading) shows that particles move; it is faster at higher temperature.

Property	Solid	Liquid	Gas
Shape	Fixed	Takes the shape of the container	No fixed shape
Volume	Fixed	Fixed	No fixed volume
Compressibility	Negligible	Very little	High
Force of attraction	Strongest	Weaker	Weakest

Change of state

- **Melting point** of ice: 273 K (0 °C). **Boiling point** of water: 373 K (100 °C). Temperature stays constant during a change of state.
- **Latent heat** is the heat used to change the state without raising the temperature (latent heat of fusion; of vaporisation).
- **Sublimation:** solid changes directly into gas, e.g. camphor, naphthalene, ammonium chloride. The reverse is called deposition.
- Increasing pressure and lowering temperature can turn a gas into a liquid. Solid CO_2 is called dry ice.

Kelvin = °C + 273 (more exactly 273.15)
Evaporation happens at any temperature below the boiling point, at the surface. It increases with surface area, temperature and wind speed, and **decreases with humidity**. Evaporation causes **cooling**.

Common mistakes

- Confusing evaporation (surface, any temperature) with boiling (whole liquid, fixed temperature).
- Writing °K. The unit is just K.

Quick check

1. Convert 25 °C to the Kelvin scale, and 373 K to °C.

2. Why do we feel cool after sweating, and why is this slower on a humid day?